Fundamentals Chapter 5

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1a. We have that $0 \cdot \mathbf{v} = (a - a) \cdot \mathbf{v} = a \cdot \mathbf{v} - a \cdot \mathbf{v} = \mathbf{0}$.

1b. We see that $\mathbf{v} + (-1)\mathbf{v} = (1)\mathbf{v} + (-1)\mathbf{v} = (1 - 1)\mathbf{v} = 0\mathbf{v} = \mathbf{0}$. So $(-1)\mathbf{v} = -\mathbf{v}$.

1c. We have that $a \cdot 0 = a(\mathbf{v} - \mathbf{v}) = a\mathbf{v} - a\mathbf{v} = \mathbf{0}$.

1d. We get that $a\mathbf{v} = a\mathbf{u} - a\mathbf{u}$, so $a(\mathbf{v} - \mathbf{u}) = -a\mathbf{u}$. This is true if $a = 0$. If not, then $\mathbf{v} - \mathbf{u} = -\mathbf{u}$, so $\mathbf{v} = \mathbf{0}$.

2. We show that $\mathbb{F}$ is a vector space over $\mathbb{F}$ by showing all the axioms to be true:

$$1 \cdot a = a,$$

$$a(bc) = (ab)c,$$

$$a(b + c) = ab + ac,$$

$$(a + b)c = a(b + c) = ab + ac.$$

These properties follow from the axioms of a field.

3. We have that,

$$1 \cdot \mathbf{v} = (1 \cdot \mathbf{v}_1, \ldots, 1 \cdot \mathbf{v}_n) = \mathbf{v},$$

$$a(b\mathbf{v}) = (ab\mathbf{v}_1, \ldots, a\mathbf{v}_n) = (ab)\mathbf{v},$$

$$a(\mathbf{u} + \mathbf{v}) = (a\mathbf{u}_1 + a\mathbf{v}_1, \ldots, a\mathbf{u}_n + a\mathbf{v}_n) = a\mathbf{u} + a\mathbf{v},$$

$$(a + b)\mathbf{v} = (a\mathbf{v}_1 + b\mathbf{v}_1, \ldots, a\mathbf{v}_n + b\mathbf{v}_n) = a\mathbf{v} + b\mathbf{v}.$$

This holds for all $a, b \in \mathbb{R}$ or $\mathbb{Q}$.

4. A polynomial, $\mathbf{v}$, of degree $k \leq n$ can be written as

$$\mathbf{v} = \sum_{i=0}^{n} a_i x^i$$

where each a_i is the coefficient of the term of degree i, and $a_i = 0$ for all $i > k$.

Now, let $a, b \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in \mathbb{F}[x]_{\leq n}$. Define the scalar multiplication of a polynomial by $a \in \mathbb{F}$ by multiplying each coefficient by a. Then the following properties are true:

$$1 \cdot \mathbf{v} = \sum_{i=0}^{n} 1 \cdot a_i x^i = \mathbf{v},$$

$$a(b\mathbf{v}) = a \sum_{i=0}^{n} b(a_i) x^i = \sum_{i=0}^{n} (ab) a_i x^i = (ab)\mathbf{v},$$

$$a(\mathbf{u} + \mathbf{v}) = a \sum_{i=0}^{n} (a_i + b_i) x^i = \sum_{i=0}^{n} (a(a_i) + a(b_i)) x^i = a\mathbf{u} + a\mathbf{v},$$

$$(a + b)\mathbf{v} = \sum_{i=0}^{n} (a + b) a_i x^i = \sum_{i=0}^{n} a(a_i) x^i + \sum_{i=0}^{n} b(a_i) x^i = a\mathbf{v} + b\mathbf{v}.$$

So $\mathbb{F}[x]_{\leq n}$ is a vector space over $\mathbb{F}$.

Also, $\mathbb{F}[x]_{=n}$ is not a vector space under the same operation because there are certain polynomials of degree n that, when added together, do not produce another polynomial of degree n, so it doesn't satisfy closure.

5. Since $\mathbb{F}[x] \subseteq \mathbb{F}[x, y]$, we can just prove it for $\mathbb{F}[x, y]$. We can denote a polynomial in x, y as

$$\mathbf{v} = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} a_{ij} x^i y^j$$

where $a_{ij} = 0$ if $i + j$ is greater than the degree of $\mathbf{v}$. We see that

$$1 \cdot \mathbf{v} = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} 1 \cdot a_{ij} x^i y^j = \mathbf{v},$$

$$a(b\mathbf{v}) = a \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} b \cdot a_{ij} x^i y^j = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} ab \cdot a_{ij} x^i y^j = (ab)\mathbf{v},$$

$$a(\mathbf{u} + \mathbf{v}) = a \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} (a_{ij} + b_{ij}) x^i y^j = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} (a(a_{ij}) + a(b_{ij})) x^i y^j = a\mathbf{u} + a\mathbf{v},$$

$$(a + b)\mathbf{v} = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} (a + b) a_{ij} x^i y^j = a\mathbf{v} + b\mathbf{v}.$$

So $\mathbb{F}[x, y]$ and $\mathbb{F}[x]$ are vector spaces over $\mathbb{F}$.

6. Define scalar multiplication of $c \in \mathbb{F}$ with a polynomial in $F[[x]]$ by

$$c \cdot \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} (c \cdot a_k) x^k.$$

Then $\mathbb{F}[[x]]$ is an abelian group over $\mathbb{F}$.

8a. Let $\mathbf{v}_1 = (1, 0)$ and $\mathbf{v}_2 = (2, 0)$. Then $\{\mathbf{v}_1, \mathbf{v}_2\}$ is not linearly independent.

8b. Let $\mathbf{v}_1 = (1, 0)$ and $\mathbf{v}_2 = (2, 0)$. Then $\{\mathbf{v}_1, \mathbf{v}_2\}$ doesn't span $\mathbb{R}^2$.